

# Cross-Lingual Dense Retrieval Using Multilingual Transformer Embeddings

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**Abstract**—Traditional information retrieval systems are constrained by monolingual search capabilities, limiting users’ access to relevant information available in other languages. This paper presents a cross-lingual dense retrieval system that enables semantic document search across English, Spanish, and French without relying on explicit translation. The system leverages the Language-agnostic BERT Sentence Embedding (LaBSE) model to represent documents in a shared multilingual semantic space and employs FAISS vector indexing for efficient similarity search. Experimental evaluation on a corpus of 5,036 Wikipedia articles demonstrates that the proposed dense retrieval approach achieves a Mean Reciprocal Rank (MRR) of 0.68 and Recall@10 of 0.82, outperforming the traditional BM25 baseline by 51% on MRR and a translate-then-retrieve pipeline by 31%. The system achieves sub-second query latency (20 ms end-to-end on CPU), making it suitable for interactive deployment. An interactive web-based demonstration validates the practical applicability of multilingual semantic search in real-world information access scenarios.

**Index Terms**—cross-lingual information retrieval, multilingual embeddings, dense retrieval, transformer models, semantic search, FAISS, LaBSE

## I. INTRODUCTION

In an increasingly globalised digital landscape, retrieving information across language barriers has become critical for researchers, businesses, and knowledge workers. Conventional information retrieval (IR) systems operate within monolingual contexts, requiring users to formulate queries in the same language as the document collection. This constraint creates significant barriers: valuable knowledge published in other languages remains undiscovered by users who lack the corresponding language proficiency.

### A. Problem Statement

Consider a researcher investigating climate-change mitigation who queries in English. A monolingual search engine returns only English documents, potentially missing high-impact work published in Spanish or French. Machine translation offers a partial remedy, yet the *translate-then-retrieve* pipeline suffers from compounding failure modes: translation errors propagate to retrieval errors, latency increases, and semantic nuances may be lost.

The fundamental challenge is therefore to enable semantic similarity matching across languages *without* relying on

explicit translation, allowing users to query in any supported language and retrieve semantically relevant documents regardless of the document’s source language.

### B. Proposed Approach

We present a cross-lingual dense retrieval system that encodes documents from multiple languages into a shared semantic vector space using the LaBSE model [5]. Semantic similarity is computed directly in this shared space via inner-product search, eliminating the need for translation at query time. Efficient retrieval is achieved through the FAISS library [6].

### C. Contributions

The main contributions of this work are:

- 1) An end-to-end cross-lingual retrieval pipeline supporting English, Spanish, and French documents;
- 2) Empirical evaluation demonstrating a 51% MRR improvement over BM25 and 31% over a translate-then-retrieve baseline on a multilingual Wikipedia corpus;
- 3) Deployment of an interactive Streamlit-based demonstration supporting real-time cross-lingual queries;
- 4) Comprehensive per-language performance analysis revealing consistent, language-agnostic retrieval quality;
- 5) An open-source implementation enabling reproducibility and straightforward extension to additional languages.

## II. RELATED WORK

### A. Traditional Cross-Lingual Information Retrieval

Early cross-lingual IR (CLIR) relied on bilingual dictionaries and parallel corpora to translate either queries or documents before retrieval [1]. These approaches are brittle in the presence of polysemy, domain-specific terminology, and low-resource language pairs where parallel data is scarce.

### B. Neural Multilingual Representations

Multilingual BERT (mBERT) [2] demonstrated that a single transformer model trained on concatenated multilingual corpora develops emergent cross-lingual transfer. XLM-RoBERTa [3] extended this with improved pre-training objectives and scale. LaBSE [5] further specialises in sentence-level embeddings by training on translation-ranking tasks over 109

[System architecture diagram: Wikipedia API → Preprocessing → LaBSE Encoder → FAISS Index → Ranked Results]

Fig. 1. High-level architecture of the cross-lingual dense retrieval system.

languages, making it particularly well-suited to cross-lingual retrieval.

### C. Dense Retrieval

Dense Passage Retrieval (DPR) [4] established that bi-encoder architectures producing dense representations consistently outperform BM25 for open-domain question answering. Sentence-BERT [7] and the Multilingual Universal Sentence Encoder [8] further demonstrated the utility of dense embeddings for semantic search. Our work extends these ideas to the multilingual retrieval setting using LaBSE embeddings.

### D. Approximate Nearest Neighbour Search

Scalable similarity search in high-dimensional spaces is a prerequisite for practical dense retrieval. FAISS [6] provides GPU-accelerated implementations of exact and approximate nearest-neighbour algorithms that operate at billion-scale, delivering sub-millisecond search latency. BEIR [10] establishes a comprehensive benchmark for evaluating zero-shot dense retrieval across heterogeneous domains.

## III. METHODOLOGY

### A. System Architecture

The proposed system comprises five sequential stages: (1) data collection, (2) preprocessing, (3) document encoding, (4) index construction, and (5) query-time retrieval. Figure 1 provides a high-level overview.

### B. Data Collection

Wikipedia was selected as the document corpus owing to its multilingual coverage, thematic diversity, and consistent editorial quality. Using the Wikipedia API, we collected up to 2,000 articles per language (English, Spanish, French) spanning 20 seed categories: Science, Technology, History, Geography, Politics, Sports, Arts, Literature, Medicine, and Environment.

Corpus construction followed a topic-based sampling procedure: (i) select a seed topic from the predefined category list; (ii) issue a Wikipedia search query; (iii) retrieve the top-10 results; (iv) download full article content and metadata; (v) apply title-based deduplication; and (vi) repeat until the per-language target is reached.

### C. Preprocessing

Raw Wikipedia articles were processed through the following pipeline:

- **Text cleaning:** Removal of HTML tags, URLs, special characters, and markup artifacts.
- **Normalisation:** Standardisation of whitespace, quotation marks, and diacritics.
- **Quality filtering:** Exclusion of stub articles (<300 characters) and list pages (>50,000 characters).
- **Deduplication:** Removal of near-duplicate articles based on title similarity.

After preprocessing, 5,036 articles were retained—1,668 English, 1,715 Spanish, and 1,653 French—representing an 84% retention rate.

### D. Document Encoding

Documents were encoded using LaBSE [5], a transformer-based sentence encoder pre-trained on 109 languages with a dual-encoder architecture optimised for translation-ranking. LaBSE produces 768-dimensional  $\ell_2$ -normalised dense vectors that preserve semantic equivalence across languages.

Each document was represented by concatenating its title and the leading summary paragraph before encoding, capturing both the topical label and the core semantic content. Encoding was performed in batches of 64 documents; GPU acceleration (NVIDIA Tesla T4) reduced the total encoding time from approximately 6 hours on CPU to 38 minutes.

### E. Index Construction

We used FAISS `IndexFlatIP` (Inner Product) for exact cosine similarity search. Because  $\ell_2$ -normalised vectors satisfy  $\cos(\mathbf{u}, \mathbf{v}) = \mathbf{u} \cdot \mathbf{v}$ , inner-product search is equivalent to cosine similarity search. The flat index ensures exact nearest-neighbour results; given the corpus size of  $\sim 5\text{K}$  documents, exact search is feasible with sub-second latency without resorting to approximate methods.

### F. Query Processing and Retrieval

At query time, the input query is encoded by the same LaBSE model into a 768-dimensional vector. FAISS then returns the top- $k$  documents (default  $k = 10$ ) ranked by cosine similarity.

### G. Baseline Methods

1) *BM25 (Monolingual)*: The Okapi BM25 ranking function [?] is applied with standard parameters  $k_1 = 1.5$ ,  $b = 0.75$ . Each language sub-collection is indexed and searched independently; monolingual queries are matched against the corresponding sub-index.

2) *Translate-Then-Retrieve*: Queries are translated into all supported languages via the Google Translate API. BM25 is then applied in each language sub-index, and results are merged and re-ranked by normalised BM25 score.

TABLE I  
OVERALL RETRIEVAL PERFORMANCE

| Method                                          | MRR         | Recall@10   | nDCG@10     |
|-------------------------------------------------|-------------|-------------|-------------|
| BM25 (monolingual)                              | 0.45        | 0.62        | 0.58        |
| Translate-then-Retrieve                         | 0.52        | 0.68        | 0.64        |
| Dense Retrieval (LaBSE)                         | <b>0.68</b> | <b>0.82</b> | <b>0.76</b> |
| <i>Relative improvement of dense retrieval:</i> |             |             |             |
| vs. BM25                                        | +51%        | +32%        | +31%        |
| vs. Translate-Retrieve                          | +31%        | +21%        | +19%        |

TABLE II  
PER-LANGUAGE MRR PERFORMANCE

| Method                  | English     | Spanish     | French      |
|-------------------------|-------------|-------------|-------------|
| BM25 (monolingual)      | 0.47        | 0.44        | 0.44        |
| Translate-then-Retrieve | 0.54        | 0.51        | 0.50        |
| Dense Retrieval (LaBSE) | <b>0.69</b> | <b>0.68</b> | <b>0.68</b> |

## H. Evaluation Protocol

1) *Test Set Construction*: Test queries were generated from 500 held-out articles by using (i) the document title and (ii) the first sentence of the summary as separate queries, yielding ground truth relevance pairs for automated evaluation.

2) *Metrics*: **Mean Reciprocal Rank (MRR)**:

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q^*} \quad (1)$$

where  $\text{rank}_q^*$  is the position of the first relevant document for query  $q$ .

**Recall@ $k$** : The fraction of queries for which at least one relevant document appears in the top- $k$  results.

**nDCG@ $k$** : Normalised Discounted Cumulative Gain, which rewards relevant documents appearing at higher ranks:

$$\text{nDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k}, \quad \text{DCG@}k = \sum_{i=1}^k \frac{\text{rel}_i}{\log_2(i+1)} \quad (2)$$

## IV. EXPERIMENTAL RESULTS

### A. Overall Retrieval Performance

Table I summarises the comparative performance of all methods on the full cross-lingual retrieval task.

The LaBSE dense retrieval system achieves substantial and consistent improvements across all three metrics, demonstrating the effectiveness of shared multilingual semantic representations.

### B. Per-Language Analysis

Table II presents MRR disaggregated by query language.

The variance across languages is negligible ( $\leq 0.01$  MRR), confirming that the LaBSE embedding space is genuinely language-agnostic for the supported language set. BM25 shows a minor English advantage (0.47 vs. 0.44), attributable to the larger English training vocabulary in the tokeniser.

TABLE III  
QUERY LATENCY BREAKDOWN (CPU)

| Operation                 | Latency (ms) |
|---------------------------|--------------|
| Query encoding (LaBSE)    | 15           |
| FAISS search (5,036 docs) | 2            |
| Result formatting         | 3            |
| <b>Total</b>              | <b>20</b>    |

TABLE IV  
QUALITATIVE RETRIEVAL EXAMPLES

| Query (EN)                     | Rank | Retrieved Document             |
|--------------------------------|------|--------------------------------|
| <i>climate change</i>          | 1    | Medio ambiente natural (ES)    |
|                                | 2    | Cambio climático (ES)          |
|                                | 3    | Réchauffement climatique (FR)  |
| <i>artificial intelligence</i> | 1    | Inteligencia artificial (ES)   |
|                                | 2    | Intelligence artificielle (FR) |
|                                | 3    | Machine learning (EN)          |

### C. Query Latency

Table III details end-to-end query latency measured on CPU.

The 20 ms total latency is well within the 100 ms threshold conventionally associated with “instantaneous” user perception, confirming suitability for interactive deployment without GPU acceleration at query time.

### D. Qualitative Analysis

Table IV illustrates representative retrieval outputs for two English queries, demonstrating successful cross-lingual semantic matching.

These examples illustrate that the system retrieves semantically equivalent concepts across languages without requiring any translation step, including topically adjacent documents (e.g., “Machine learning” as a sub-discipline of AI).

## V. DISCUSSION

### A. Key Findings

**Dense retrieval superiority.** The 31% MRR gain over the translate-then-retrieve baseline demonstrates that multilingual transformer embeddings surpass explicit translation pipelines. We attribute this to two factors: (i) translation errors do not compound with retrieval errors, and (ii) LaBSE encodes pragmatic and contextual meaning that may be partially lost in literal translation.

**Language-agnostic performance.** The near-identical MRR across English (0.69), Spanish (0.68), and French (0.68) indicates balanced cross-lingual capability, an important practical property for equitable information access.

**Practical efficiency.** Offline corpus encoding benefits from GPU acceleration ( $6\times$  speedup), while online query serving requires no GPU, operating at 20 ms on commodity hardware.

### B. Limitations

**Evaluation methodology.** Test queries derived from document titles and leading sentences may not reflect the full

TABLE V  
KEY SOFTWARE DEPENDENCIES

| Library               | Version | Purpose                  |
|-----------------------|---------|--------------------------|
| Sentence-Transformers | 2.2.0   | LaBSE inference          |
| FAISS                 | 1.7.4   | Vector indexing & search |
| PyTorch               | 2.0.0   | Neural network backend   |
| Streamlit             | 1.25.0  | Web demonstration        |
| pandas / NumPy        | –       | Data manipulation        |

distribution of real user information needs. Future evaluation should leverage search logs or crowdsourced relevance judgements.

**Corpus scale.** The 5,036-document corpus constitutes a proof-of-concept. Production-scale deployment ( $\geq 1\text{M}$  documents) would require approximate nearest-neighbour indices (e.g., FAISS IVF with product quantisation) and a re-ranking stage to maintain precision.

**Language coverage.** The current system is limited to three high-resource Indo-European languages. Performance on morphologically rich or typologically distant languages (e.g., Arabic, Finnish, Japanese) with lower LaBSE training coverage remains unexplored.

**Domain specificity.** Wikipedia’s encyclopaedic register may not generalise to specialised domains such as biomedicine or legal text, where domain-adaptive fine-tuning of the encoder may be necessary.

### C. Future Work

Several directions merit investigation:

- 1) **Hybrid retrieval:** Interpolating dense and sparse (BM25) scores through learned late fusion to exploit complementary strengths;
- 2) **Cross-encoder re-ranking:** Applying a multilingual cross-encoder as a second-stage ranker to refine top- $k$  candidates;
- 3) **Query expansion:** Incorporating pseudo-relevance feedback or query reformulation to improve recall on under-specified queries;
- 4) **Language extension:** Evaluating on Arabic, Chinese, Hindi, and Swahili to assess coverage of LaBSE on lower-resource and non-Indo-European languages;
- 5) **User studies:** Conducting task-based evaluations with real users to measure utility beyond automated IR metrics.

## VI. IMPLEMENTATION AND DEPLOYMENT

### A. Software Stack

The system is implemented in Python 3.8 using the libraries listed in Table V.

The codebase is organised into modular components for data collection, preprocessing, document encoding, indexing, retrieval, and evaluation, facilitating independent development and testing of each stage.

### B. Interactive Web Demonstration

A Streamlit-based web interface enables real-time demonstration. Users may enter a query in any supported language and receive top-10 results annotated with language flags, cosine similarity scores, and expandable document summaries. A sidebar displays system statistics including index size, encoding time, and per-query latency.

### C. Reproducibility

The complete source code, pre-built FAISS index, and evaluation scripts are released as open-source software. A README provides step-by-step setup instructions, configuration files, and usage examples to facilitate independent reproduction and extension.

## VII. CONCLUSION

This paper presented a cross-lingual dense retrieval system that enables semantic document search across English, Spanish, and French without explicit translation. By representing documents from all languages in a shared LaBSE embedding space and coupling this with FAISS similarity search, the system achieves  $\text{MRR} = 0.68$  and  $\text{Recall}@10 = 0.82$ —a 51% MRR improvement over BM25 and a 31% improvement over a translate-then-retrieve pipeline.

The language-agnostic embedding space is the core enabler: it eliminates translation overhead, prevents error compounding, and preserves semantic nuance. Consistent per-language performance and 20 ms CPU query latency confirm both the effectiveness and the practical deployability of the approach.

Future work will scale the system to larger, domain-diverse corpora, extend coverage to lower-resource languages, investigate hybrid dense–sparse retrieval, and conduct user studies to evaluate real-world utility. As multilingual information needs continue to grow, cross-lingual retrieval systems represent a foundational technology for democratising equitable access to global knowledge.

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